

Lecture 8: I/O & Storage devices

CS211 - Operating systems

Spring 2026



I/O devices

???

I/O devices

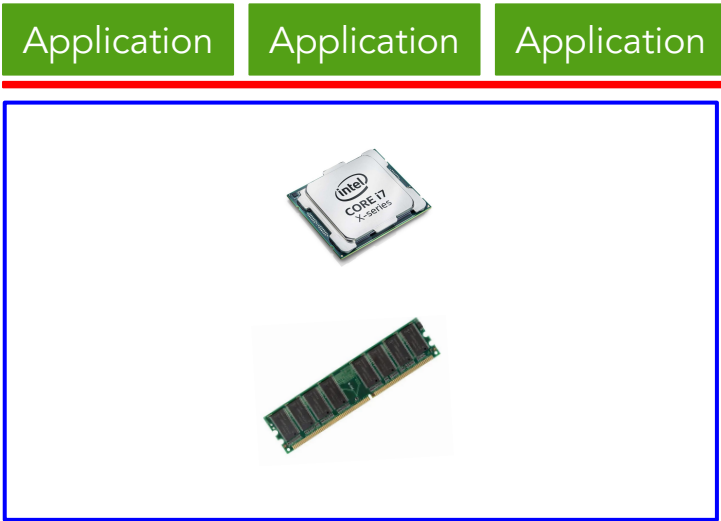


Standard interfaces

- Block Devices (e.g. disk drives, tape drives, DVD-ROM)
 - Access blocks of data
 - Commands include `open()`, `read()`, `write()`, `seek()`
- Character Devices (e.g. monitor, keyboards, mice, serial ports, some USB devices)
 - Single characters at a time
 - Commands include `get()`, `put()`
- Network Devices (e.g. Ethernet, Wireless, Bluetooth)
 - Unix and Windows include socket interface

I/O layer

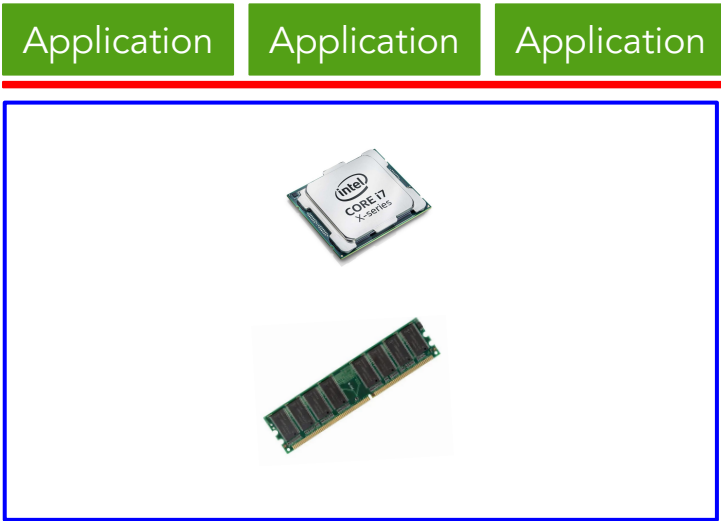
OS
interface



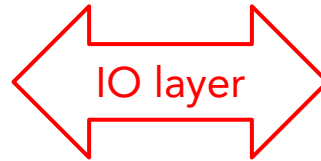
Virtual Machine Abstraction

I/O layer

OS
interface



Virtual Machine Abstraction



I/O devices

I/O layer requirements

- Thousands of devices, each slightly different
 - OS: How can we standardize the interfaces to these devices?
- Devices unreliable: media failures and transmission errors
 - OS: How can we make them reliable???
- Devices unpredictable and/or slow
 - OS: How can we manage them if we don't know what they will do or how they will perform?

Example

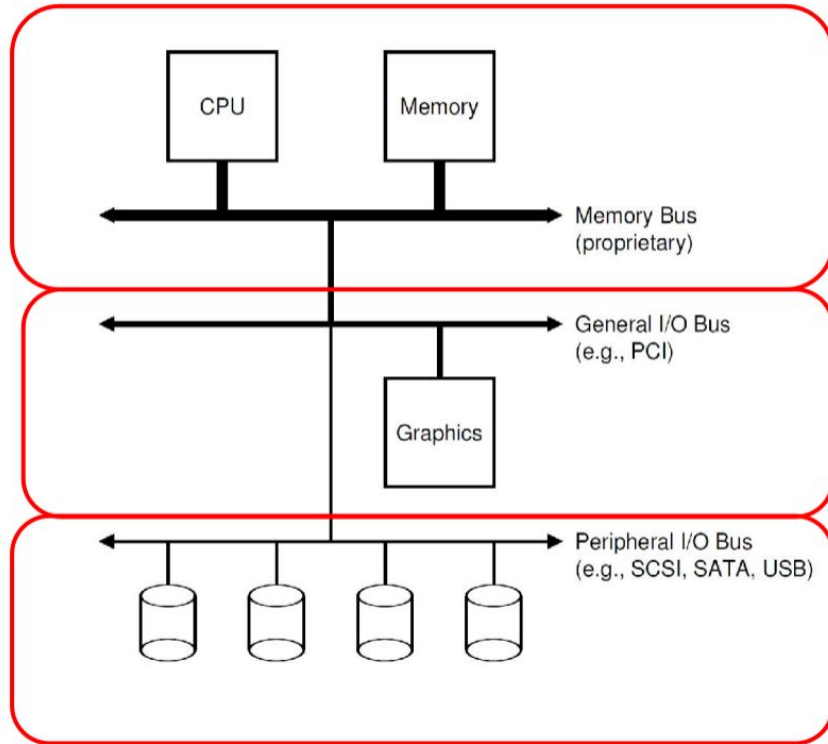
```
FILE fd = fopen("/dir/file.txt", "rw");  
for (int i = 0; i < 10; i++) {  
    fprintf(fd, "count %d\n", i);  
}  
close(fd);
```

Example

```
FILE fd = fopen("/dir/file.txt", "rw");  
for (int i = 0; i < 10; i++) {  
    fprintf(fd, "count %d\n", i);  
}  
close(fd);
```

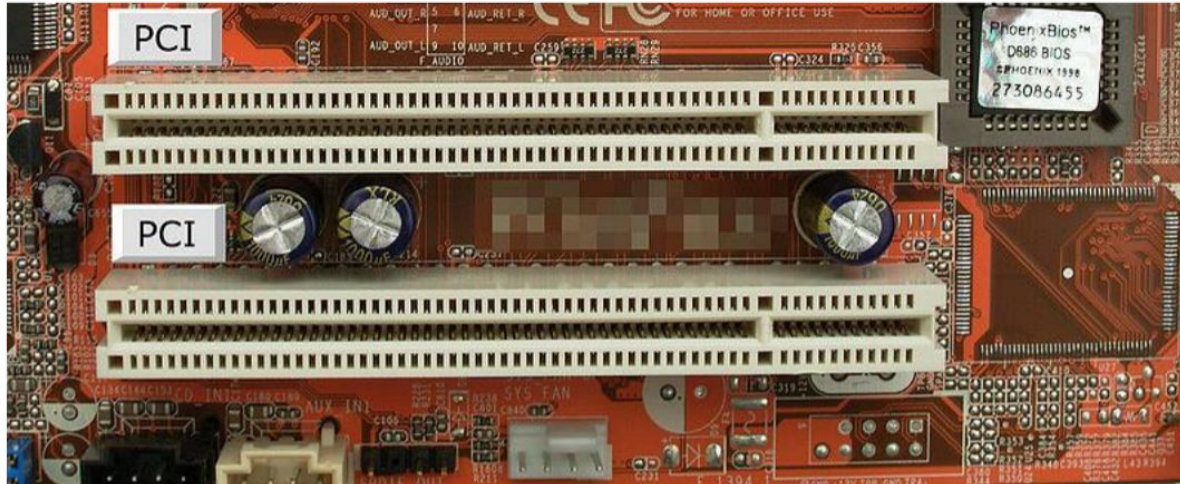
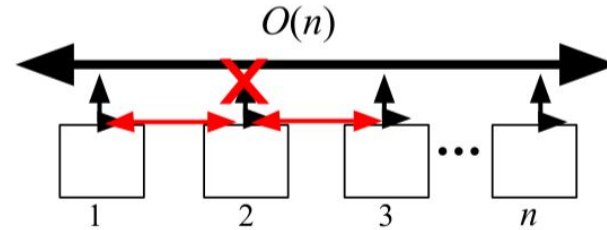
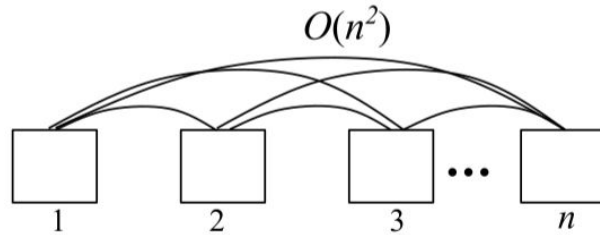
This applies for any files stored in any storage devices

I/O architecture



Why a bus?

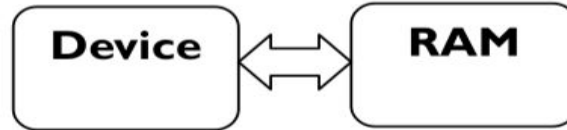
Why a bus?



How does the processor talk to devices?

- Interrupts
- Polling
 - Report results through status register that processor looks at periodically

Programmed IO vs Direct memory access



Device drivers

➤ Device Driver

- Device-specific code in the kernel that interacts directly with the device hardware
- Implements the OS abstraction

➤ The OS is 70% device-drivers

Storage devices



IBM 350 (1956): 3.75 MB



SSD vs HDD

	SSD vs. HDD Usually 10,000 or 15,000 rpm SAS drives	
0.1 ms	Access Times SSDs exhibit virtually no access time	5.5-8.0 ms
SSDs deliver at least 6000 io/s	Random I/O Performance SSDs are at least 15 times faster than HDDs	HDDs reach up to 400 io/s
SSDs have a failure rate of less than 0.5%	Reliability This makes SSDs 4-10 times more reliable	HDDs failure rate fluctuates between 2-5%
SSDs consume between 2 and 5 watts	Energy Savings This means that on a large server, approximately 100 watts are saved	HDDs consume between 6 and 15 watts
SSDs have an average I/O wait of 1%	CPU Power You will have an extra 6% of CPU power for other operations	HDDs average I/O wait is about 7%
The average service time for an I/O request while running a backup remain below 20 ms	Input/Output Request Times SSDs allow for much faster data access	The I/O request time with HDDs during backup rises up to 400-500 ms
SSD backups take about 6 hours	Backup Rates SSDs allow for 3-5 times faster backup for your data	HDD backups take up to 20-24 hours

... still keep updating ...

Article | [Open access](#) | Published: 18 February 2026

Laser writing in glass for dense, fast and efficient archival data storage

[Microsoft Research Project Silica Team](#)

[Nature](#) 650, 606–612 (2026) | [Cite this article](#)

92k Accesses | 5 Citations | 1109 Altmetric | [Metrics](#)

Abstract

Long-term preservation of digital information is vital for safeguarding the knowledge of humanity for future generations. Existing archival storage solutions, such as magnetic tapes and hard disk drives, suffer from limited media lifespans that render them unsuitable for long-term data retention^{1,2,3}. Optical storage approaches, particularly laser writing in robust media such as glass, have emerged as promising alternatives with the potential for increased longevity. Previous work^{4,5,6,7,8,9,10,11,12,13,14,15,16} has predominantly optimized individual aspects such as data density but has not demonstrated an end-to-end system, including writing, storing and retrieving information. Here we report an optical archival storage technology based on femtosecond laser direct writing in glass that addresses the practical demands of archival storage, which we call Silica. We achieve a data density of $1.59 \text{ Gbit mm}^{-3}$ in 301 layers for a capacity of 4.8 TB in a 120 mm square, 2 mm thick piece of glass. The demonstrated write regimes enable a write throughput of 25.6 Mbit s^{-1} per beam, limited by

Data stripping



Disk 1

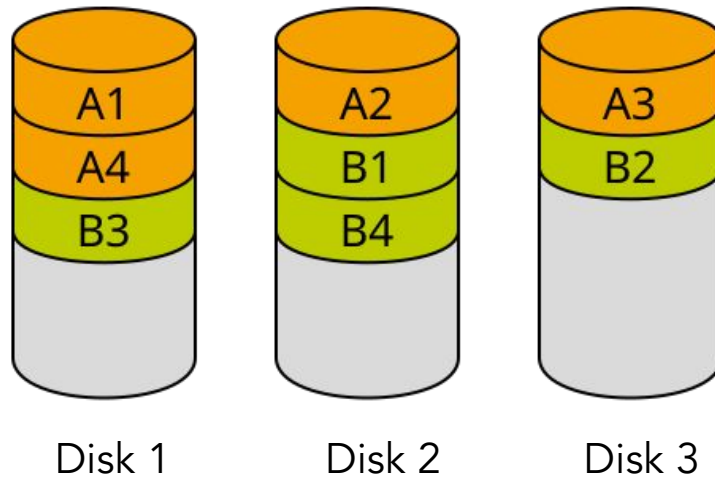


Disk 2

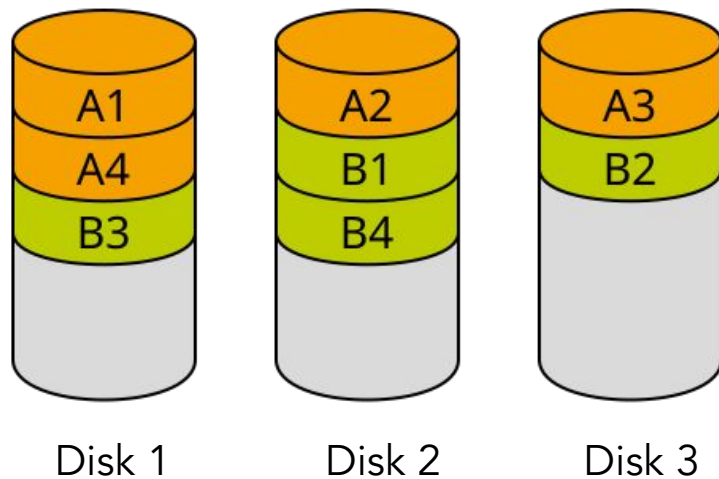


Disk 3

Data stripping



Data stripping



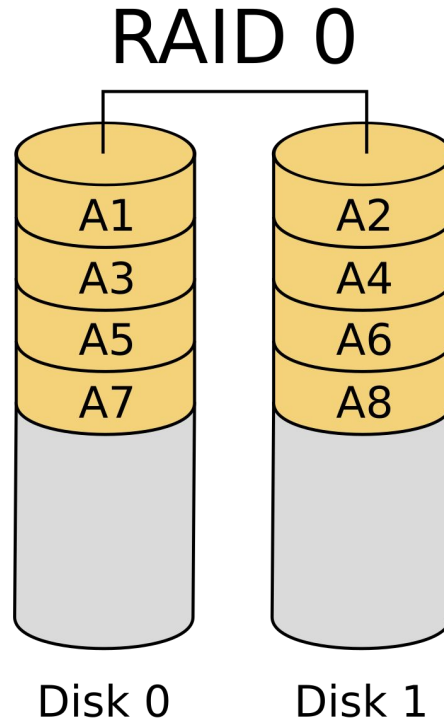
Why is it better?

- Concurrency
- Load-balancing

RAID

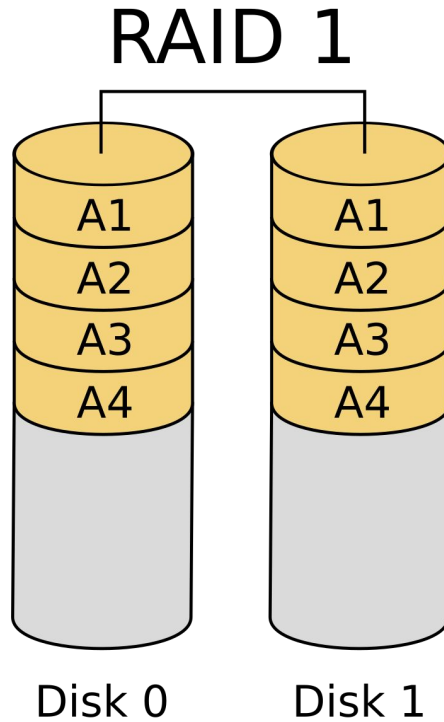
- "redundant array of inexpensive disks"
- or "redundant array of independent disks"

RAID 0



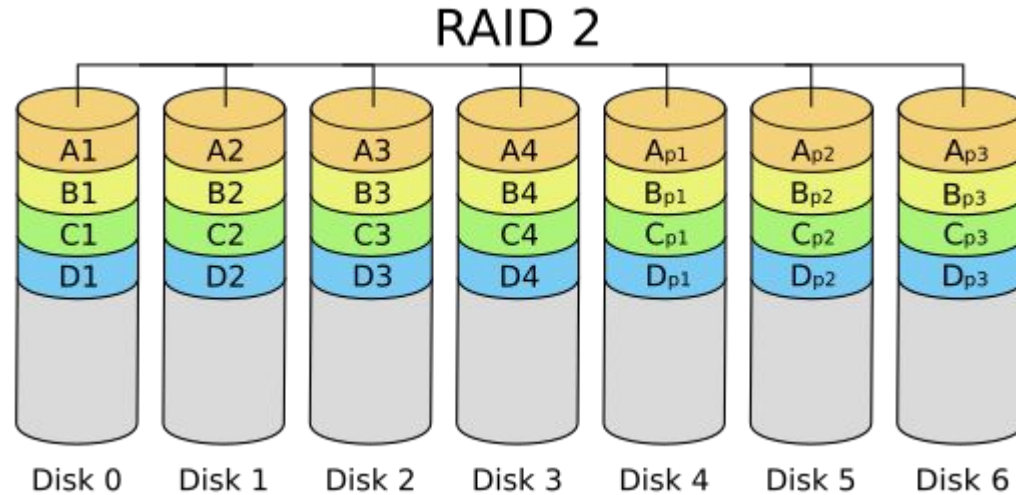
Speed?
Reliability?
Cost?

RAID 1



Speed?
Reliability?
Cost?

RAID 2

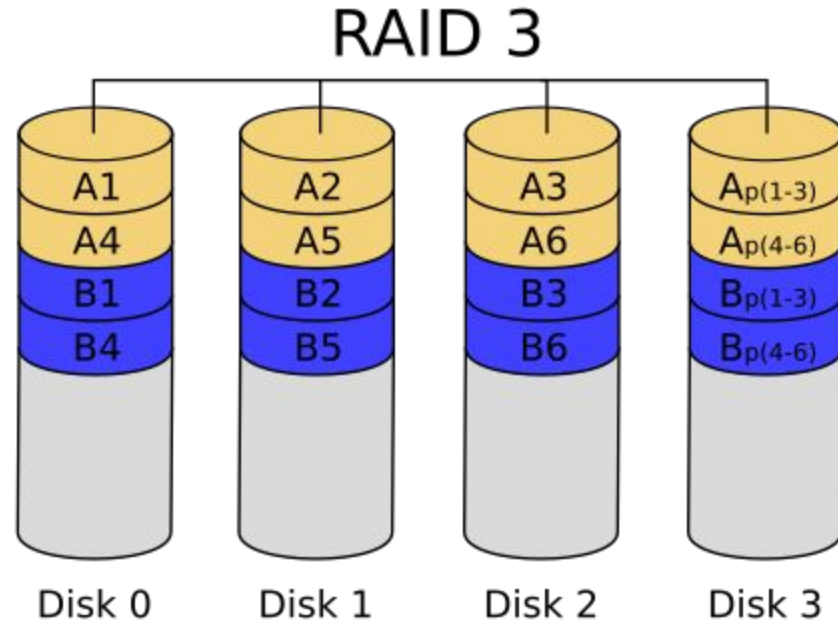


Speed?
Reliability?
Cost?

$$D_N = D_1 \oplus D_2 \oplus \dots \oplus D_{N-1}$$

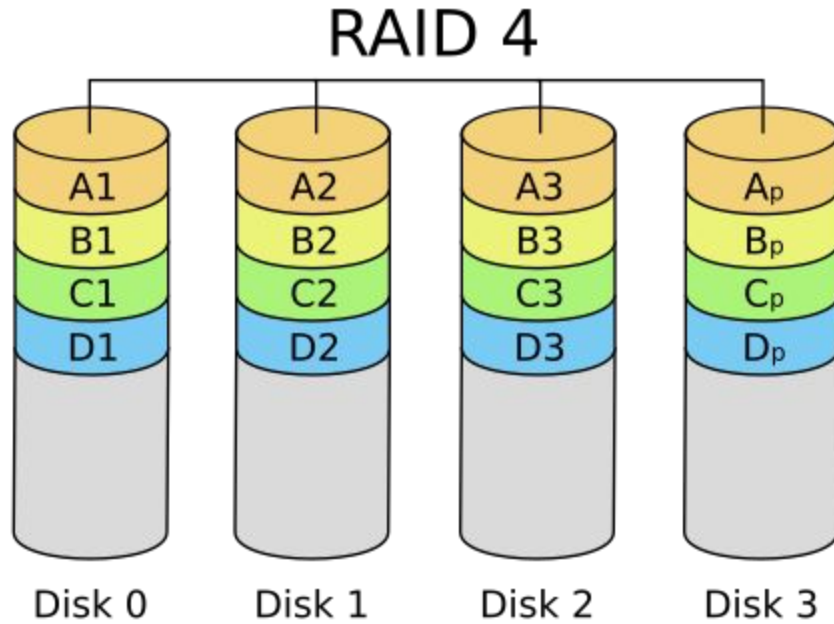
- $\oplus$ = XOR operation

RAID 3



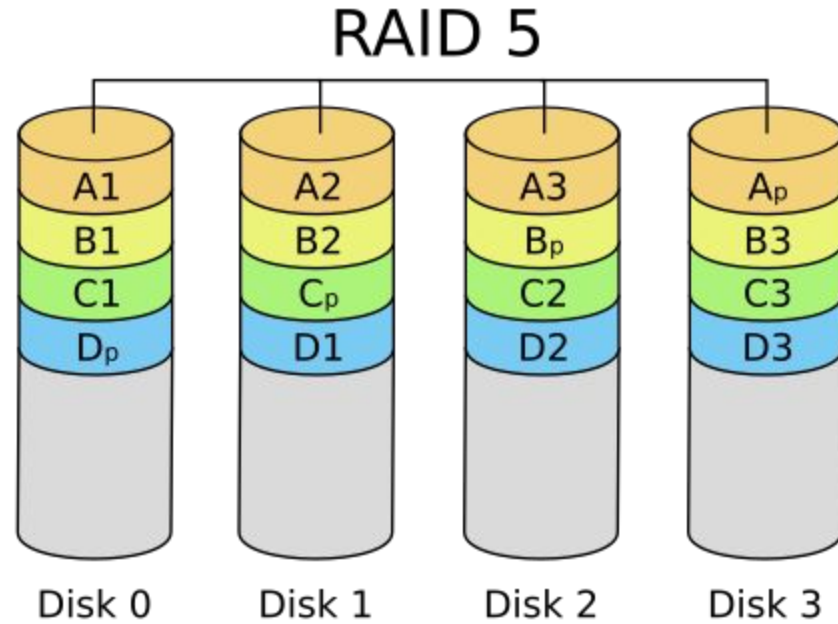
Speed?
Reliability?
Cost?

RAID 4



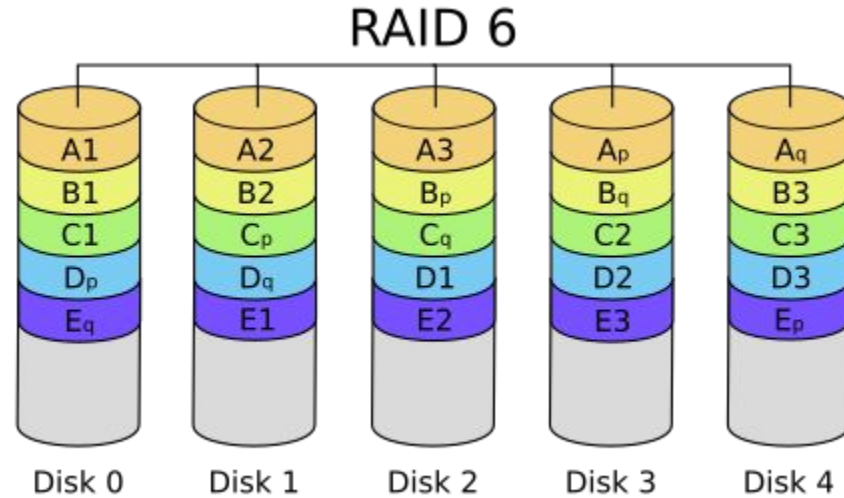
Speed?
Reliability?
Cost?

RAID 5



Speed?
Reliability?
Cost?

RAID 6



Speed?
Reliability?
Cost?

Summary

- I/O devices
- Storage devices
- RAID